

CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 16

Name (Print): _____ Section: _____

1. Errors vs Exceptions.

Fill in the blanks:

- An _____ is a serious problem that occurs outside the program's control.
- An _____ is an unexpected event that occurs during program execution and can often be handled.

2. Examples.

Provide one example of each:

- Error: _____
- Exception: _____

3. True or False. Circle one.

- | | | |
|---|------|-------|
| • Errors are usually caused by mistakes in your program logic. | True | False |
| • Exceptions allow a program to respond to unexpected situations. | True | False |
| • A program can typically recover from an Error using try/catch. | True | False |

4. Try / Catch / Finally (Fill in the Keywords).

Complete the code below by filling in the missing Java keywords so that:

- the level file is loaded safely
- a message is printed if the file is missing
- cleanup code always executes

```
1  _____ {
2      readFile(file);
3  }
4  _____ (FileNotFoundException e) {
5      System.out.println("Level file not found.");
6  }
7  _____ {
8      System.out.println("Finished attempting to load level.");
9  }
```

5. Return Statements (Game Loading Scenario).

A game attempts to load a level and returns a status code:

- 1 = level loaded successfully
- -1 = level file not found
- 0 = cleanup completed

Consider the following method:

```

1 public static int loadGame() {
2     try {
3         loadLevel("level1.txt");
4         return 1;
5     }
6     catch (FileNotFoundException e) {
7         return -1;
8     }
9     finally {
10        System.out.println("Cleaning up resources...");
11        return 0;
12    }
13 }

```

- What value does `loadGame()` return if the level file exists?

- What value does `loadGame()` return if the level file does NOT exist?

6. Scanner Methods (Identify the Method).

Write the correct Scanner method for each description.

Description	Scanner Method
Reads the next token (word) from the input	
Reads an entire line of input	
Checks if another line exists without consuming it	

7. Using `throws` passes responsibility for handling the exception to the caller. True False

8. Before You Leave

Write one question you still have about today's topic.